

Exam Questionnaire Preparation Strategy of CSE321

Summer 2026 Version

CSE321: Operating Systems follows a common standard for both midterm and final exams questionnaire preparation based on the recommendation of the CSE 321 team where all sections' instructors participate. According to the standard, points distribution according to the difficulty levels of questions is as follows:

1. 35% questions will fall into the **easy (E)** category.
2. 35% questions will fall into the **moderate (M)** category.
3. 30% questions will fall into the **difficult (D)** category.
 - a. Among the difficult questions 10% will fall into the **advanced (A)** category.

From now on, CSE 321 exam questions will have the category markers with the questions along with CO markers in the format CO\${CO-number}\$-CA\${category-initial}.

There is an imbalance in the decomposition of mathematical and theory topics in the syllabus of the midterm and the final exams. Where there are hardly any mathematical aspects in the chapters covered before the midterm exam, the after-midterm exam syllabus allows mathematical questions from most chapters. Consequently, an even decomposition of mathematical and theory questions in exam questionnaires isn't feasible. Thus, we decided to group questions into the following classes based on their nature:

1. Theoretical Questions
2. Problem Solving Questions
 - a. Program Interpretation
 - b. Error Detection/Correction
 - c. Mathematical Calculation
3. Simulation Questions

The questionnaire in both midterm and final exams will have 35 to 50% theoretical questions and 50 to 65% questions from the other classes. Note that the number of questions and the nature of questions asked on different topics in an exam questionnaire may vary semester to semester.

What makes a question easy or difficult is often a common topic of argument, particularly for the students. Therefore, CSE321 team established a clear criterion for assessing the difficulty level of a question, which is as follows:

- **An easy question** is the one that requires only remembering things discussed in class or presented in slides. If it is a theory question then rote memorization of definitions or processes fall in this category. If it is a simulation or problem-solving question then it is very similar to examples shared in class, or slides, or some

practice sheet. Note that a lengthy simulation is lengthy, but may still be an easy question. However, understanding what is being asked in an easy question will be straightforward.

- **A moderate question** is the one that requires reading the textbook to remember the reason behind theoretical things such as pros and cons of approaches, the mechanism of implementing something, or tradeoff between implementation choices, etc. For a problem-solving question, it can be interpreting interesting algorithmic program outputs or solving computational problems that are related to practice and exercise problems in the sense that the underlying elements are common but their application requires some thoughts or more steps. A moderate simulation question may ask you to simulate the output of some OS behavior whose logic is discussed in class and explanation is available in the textbook, but a similar simulation may not be shown in class.
- Finally, **a difficult question** is the one that tests the students grasp on operating system concepts in general. If it is a theory question then the answer is directly not available to the textbook but rather derived from the textbook content. A difficult problem-solving question may ask for detecting and fixing subtle program errors, or ask to determine the consequence of having an alternative of the normative design/implementation choices presented in the textbook.